Does a machine learning approach to predicting flood defence condition grade improve on established heuristic baselines?

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Abstract

Flooding is a serious risk in England, and the EA has the responsibility of managing a large and aging network of flood defence assets, which they give condition grades in regular manual inspections. This project seeks to show that data driven solutions could be developed to assist in this process. The existing heuristics have been evaluated and ML classification methods have been developed, with the two compared. It was found that ML had a significant uplift in precision over heuristics and baselines when looking at the boundary of predicting acceptable conditions vs unacceptable, although performance was not strong enough to recommend immediate development of early warning and intervention tools following this approach.

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Respository URL: https://github.com/danielmh111/flood_defence_condition

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Chapter 1

Introduction

Flooding is a serious, rising risk in England (HM Government, 2026; National Audit Office, 2020), and maintenance of flood defence infrastructure is an important mitigation for protecting welfare of citizens and the environment. This responsibility, including scheduling of thousands of inspections a year, falls to the Environment Agency (EA). Evaluation of the condition of flood defences is done by a three tier system of visual and manual engineering inspections, needing specific experience, local knowledge, and expertise of engineers (Brown, Chatterton and Purcell, 2014). This project seeks to undertake research showing data driven approaches could improve efficacy in this task.

This project aims to use open datasets that can cover the majority of flood defences in England to explore how machine learning (ML) prediction of condition grades of flood defence assets can be done. Identification, collection, and processing of appropriate datasets is within scope, as is design, training, and evaluation of ML classifiers. Within scope, methods of prediction using existing engineering information will be explored to provide comparisons for model outcomes. The project will provide both qualitative assessment of the degree to which inspection currently relies on hard-to-access domain expertise, and direct comparison with developed data-driven approaches.

The scope excludes policy recommendations, building workings tools for screening at risk assets, or validation of results against future asset inspections. The project aims to provide quantitative evaluations of how ML models results indicate feasibility of such tools being developed in future work.

The project will determine whether existing engineering information in EA documentation is sufficient to build any form of early intervention recommendation tool, or whether new data-driven approaches are necessary. It will explore how ML approaches to predicting condition grades perform compared to methods derived from the EA's established heuristic deterioration curves, using condition grades of recent inspections, attained from the EA. This will be done in a reproducible way, hosted publicly to be repeated or extended by anyone.

This report describes background research and active research tasks carried out before presenting the results of trained ML models and evaluating the created artifact. Chapter 4 describes the exploration and implementation of the heuristic deterioration models published by the Halcrow group. This is key to describing why data science approaches are necessary, and essential context for benchmarking their performance. Appendices of the report include attributions of data usage and AI usage, and data dictionaries to describe features.

Chapter 2

Research

2.1 User Need and Rationale

Flooding in England is a significant risk, affecting citizen safety and wellbeing, damaging to the environment and incurring substantial costs, both material and economic (Ficarra and Mari, 2025; Skouralis and Lux, 2023; Beltrán, Maddison and Elliott, 2019). All forms of flooding are deemed to have significant effect by the Cabinet Office UK National Risk Register report (HM Government (2026), chapter 4 pg. 180-185), which also details the EA's role in leading efforts preparing for and responding to flooding. The EA therefore has an unwieldy task: to assess, inspect, and maintain a large and aging system of flood defences while flood risks and impacts continue to change. The AIMS register (Environment Agency, 2020) comprises 141,996 flood defence assets, of which 37,364 are maintained by the EA. Greenpeace Unearthed published an article in 2023 describing how many crucial flood defences were in an inadequate condition prior to major flooding in 2023 (Clarke, 2023). Emma Hardy MP's response to Gideon Amos MP's written questions about flood control reveal that the EA schedules around 110,000 inspections per year, potentially increase to 165,000, and additionally that around 7.1% of assets currently fall below the target grade, indicating no improvement on the 7% figure put forward in the Greenpeace Unearthed article. The question asked specifically what the actual cost was of routine maintenance and capital repair on existing flood defences, and the lack of a direct answer in Hardy's response could indicate that the EA struggles to measure these data. (Gideon Amos, 2026b, 2026c, 2026a).

Given the significant challenge that the EA faces to maintain and improve standards of flood preparedness, data-driven approaches to developing tools and processes that could improve upon or support existing measures should be explored. The artifact is a ML research pipeline that could lead to a tool meeting the above need when applied: e.g. screening assets to schedule upcoming inspections in order to direct resources towards those likely to find a deteriorated grade and therefore need further action. The user is the EA, or more generally a flood defence asset manager, with responsibility for scheduling and carrying out routine inspection and maintenance, and therefore will desire to be able to schedule inspections proactively when there is a heightened chance that the result of inspection will be a condition grade below target. As described above, the EA currently commissions inspections in three tiers, where the first tier is mostly driven by the asset type, and higher tiers are driven by escalation based on outcomes. If the user had access to a tool that allowed them to know which assets where more likely to need escalation, then they could be more flexible in how inspections were scheduled and could make more efficient use of available resources to provide better outcomes for flood defence asset health.

In order to have operational impact, the artifact will need to provide information to the user unobtainable using existing heuristic guidance. The underlying hypothesis is that, since it is trained on results of expert assessments, the ML model will encode some expert domain knowledge that trained engineers and local experts have, making it accessible to an asset manager operating at national scale. In order to assess the success of this tool, it's ability to predict asset condition grades will be compared to naïve baselines and to heuristic curves given in the Halcrow document. The project's research question is therefore: "*Does a machine learning approach to predicting flood defence condition grade improve on established heuristic baselines?*". Additionally, results must demonstrate potential to be operationally useful in order to meet user need. This means that the results can be filtered and presented so that the assets with useful findings are identified, and that confidence in predictions is sufficient to replace or complement an existing part of flood defence maintenance process.

2.2 Literature

The EA designed a comprehensive and highly involved process for inspections to produce asset grades from 1 to 5, with 1 being an asset in perfect condition and 5 an asset in very poor condition (Environment Agency, 2025 pg. 2; Halcrow Group Limited, 2013). The process combines visual inspections, engineering surveys, and specialist investigations (Brown, Chatterton and Purcell, 2014). The goal of the EA is to be proactive in planning inspections based on risk of failure, rather than reacting to events indicating failed or deteriorated asset. The design of the inspection hierarchy is intended to be efficient, allowing tier 2 and 3 inspections (engineering surveys and specialist investigations) to be sought using information presented in routine tier 1 visual inspections.

The established tools for predicting rate of deterioration of flood defence are heuristics presented in the Halcrow document (Halcrow Group Limited, 2013, pg. 4, table 1.1). These are curves and tables that describe an asset's deterioration from grade 1 to 5 based on inputs including its type, local environment, material, and maintenance regime. Specific inputs may be different for different assets (Halcrow Group Limited, 2013, pg. 6, section 2.1, table 2.1). These curves have been designed based on practitioner and asset manager experience, and while they are intended specifically for scheduling asset specific inspection and monitoring (Halcrow Group Limited, 2013, pg. 4, section 1), selecting all correct inputs for each asset explicitly requires engineering knowledge, local experience, and professional judgement (Halcrow Group Limited, 2013, pg. 9, section 2.2). The curves themselves are data informed but not statistically fitted models of deterioration rates. Because of both the required expertise and form of required inputs, this project found that heuristics do not make for effective tools for predicting a current inspection outcome for all assets, nationally, using the publicly available registers and data sets. For example, selecting the correct decay rate band would require an engineer's insight into the asset and its specific exposure in its locality (Halcrow Group Limited, 2013, sec.2.3), and data such as the width of the an embankment crest or whether it has a reinforced base are not recorded for each asset in the AIMS dataset or an equivalent register maintained by the EA (Halcrow Group Limited, 2013, pg. 8; Environment Agency, 2025 pg. 4-5). The report will show that sensible assumptions could not be found to make heuristic base prediction work at a national scale.

The prediction of infrastructure condition is an established problem that has widespread and well understood benefits, borne out by the existing literature presenting methods for more accurate, more efficient, or more timely solutions to this problem. These examples span different types of infrastructure and environments, but also share common themes. In many cases, ML has been identified as an appropriate technique for taking an existing register of infrastructure or an open dataset and trying to predict the rate of deterioration, the current condition, or the probability of the condition state changing.

Fard and Fard (2024) explored the use of a large dataset to predict the condition of the bridge decks in the USA, and employed ML models such as Random Forest, XGBoost, and ANNs. They found overall accuracies between 79.4% and 83.4% could be achieved, with F1 scores between 0.775 and 0.797. The motivation of this paper is similar to this project - predicting the condition grade could improve the efficacy of maintenance and repair planning. The scope of this project was national, as is the flood defence asset problem, although the National Bridge Inventory (their core dataset) comprises 319404 bridge decks - significantly larger than the usable AIMS data. Fard and Fard augmented their national register of infrastructure assets with multiple additional sources - traffic, climate - and used geospatial information science (GIS) and remote sensor sources to compile this. This is comparable to the collection of flood exposure, geological, climate, and geometry data from national archives and GIS sources for this project. Fard and Fard explored the relative importance of the different sources using permutation based variable importance techniques, whereas this project ueses SHAP analysis scores.

Another paper focusing on a similar problem, also concerned with the condition state of bridges, was written by Mia and Kameshwar to explore how statistical methods and large datasets could be applied as an alternative to manually evaluated inspection reports (Mia and Kameshwar, 2023). This research also used the NBI dataset and achieved accuracy over 90%. Relevant to this flood defence asset project, this paper discusses the impact that a skewed dataset with a majority condition grade can have on ML models and how biases that arise from this can be mitigated.

Both of these projects use a high quality central dataset. NBI data is highly complete, and so any bridge with any missing field could be discarded and leave enough data for adequate training and testing to prove the efficacy of ML (Mia and Kameshwar, 2023 pg. 5 section 2.1). In particular, the age of the asset and proof of yearly inspections could be established. This is a point of difference from data available for the flood defence research, where the AIMS dataset seemed to indicated that the age variable was unreliable in some cases, and the data provided by EA via EIR highly redacted.

There is also existing research directly related to flood defence condition estimation. Tarrant et al. seeks to identify damage and weakness in flood defences by making use of remote sensor data. In particular, they make use of Light Detection and Ranging (LIDAR) data to try and identify specific damage to embankments that could lower the condition grade (Tarrant *et al.*, 2017). The paper establishes two points important for this project: that flood defence grades can be driven by mechanisms such as ground movement; and that LiDAR data is one way that data can be collected for individual assets. A key difference is that the Tarrant research is done at an individual asset scale, making use of high resolution data. This project seeks to apply some of those learnings to all assets at a national scale.

Chapter 3

Proposed Artefact

The proposed artefact is a reproducible ML pipeline. It will include source code for feature extraction, infrastructure for running experiments and producing metrics, and reusable notebooks for evaluating results. This will allow practitioners and researchers to inspect the work, quickly build the ML models, and integrate further work.

ML models researched in this pipeline will be classifiers for predicting condition grade of flood defence assets.

The pipeline will be hosted publicly to make research and results accessible, and will contain environment lock file for python and R to ensure future collaborators can reuse all code (Hill, n.d.). Data will be tracked using DVC (data version control) but not hosted in github due to size. This allows simple data sharing including intermediate and processed files in future.

ML models will be presented in the context of how they could become useful to practitioners managing flood defences. A tool is envisioned to flag deteriorating assets for planning intervention.

Research work using the AIMS dataset is limited, and applications of ML to this dataset at national scale has not been done based on reviewed literature, meaning this artefact can make a novel contribution to the field while building upon existing material in the Halcrow group report, the field of infrastructure deterioration research, and the findings of Tarrant *et al.* (2017) on detecting signs of weakness in flood defences.

Chapter 4

Heuristic Investigation

4.1 Individual Heuristic Investigation

4.1.1 Rationale

The motivation for the artefact is to investigate advantages ML methods have over existing processes, which make use of heuristics curves and tables detailed in the Halcrow document. In order to use these as a comparison, a research task was designed to faithfully use heuristics to produce high quality predictions. The rationale has several parts:

- Understand specific tasks necessary to make a faithful prediction of condition grade using halcrow information
- Qualitatively understand feasibility of creating a national scale screening tool using halcrow predictions
- Measure performance of heuristic predictions for use as a comparative benchmark when evaluating ML approaches
- Qualitatively understand quantity and depth of expert knowledge required to use halcrow information effectively

It is acknowledged that the author's lack of expertise in flood defence engineering is a major caveat to any predictions made. This is explicitly specified in Halcrow guidance and means results presented here should not be interpreted as the outcome of perfect application. However, undertaking the task with best intentions to learn and apply good judgement will lead to better understanding of what and how much expertise is needed. This speaks directly to the project aims - to provide a way a non-expert responsible for the entire national register of flood defences could benefit from ML tools encoding that expertise.

4.1.2 Method

The investigation method is detailed in Figure 4.1. This describes the multi-step process and details actions taken and decisions made to produce predictions. Since it was anticipated this would be intensive and open ended, this was a time-boxed task and initial data was strictly scoped to a manageable subset of the flood defence register at the outset. Where there was any doubt about the outcome of a process, such as material of an asset, the asset was rejected. The focus of this task was to ensure there was enough data to use heuristics without making assumptions about asset characteristics. The final dataset contained only 11 high-confidence assets that were used to make predictions after extensive search of public information sources. All collected data are available in the code repository.

Once collected, heuristic curves were implemented in a python notebook to predict an asset's current condition grade. Simple metrics were calculated to evaluate success of the method. The implementation is in notebook 13.

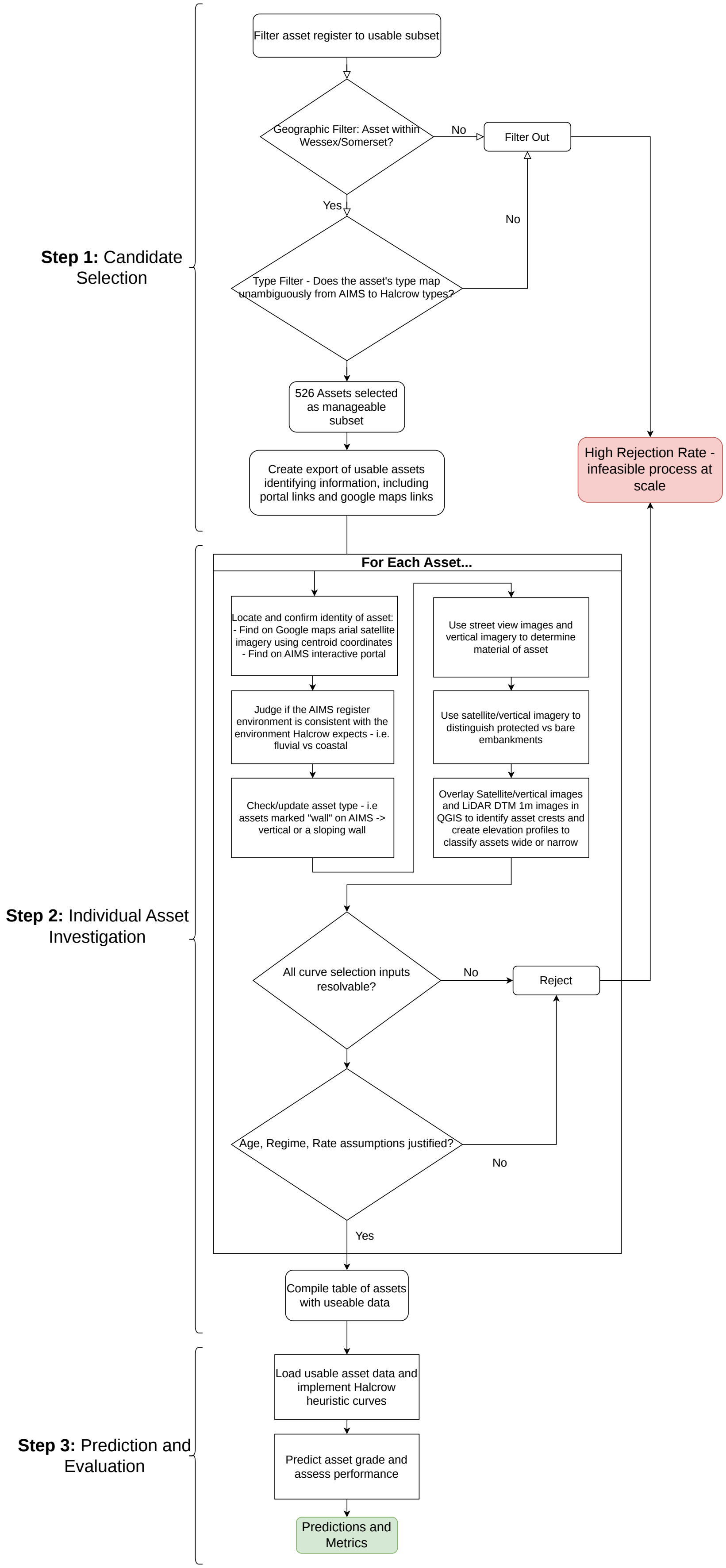

Figure 4.1: Individual Asset Heuristic Investigation

4.1.3 Results and Findings

Table 4.1: Heuristic Prediction Metrics

	Assets Evaluated	MAE (Discrete ordinal grade)
Individually Assessed Heuristic	11	0.272727
Individually Assessed Heuristic - Embankments Only	5	0.6
National Scale Heuristic	14262	0.794041
Naive Median Prediction	11	0.636364
Naive Median Prediction - Embankments Only	5	0.2
Naive Distribution Prediction	11	1.272727
Naive Distribution Prediction - Embankments Only	5	0.8

Table 4.2: Heuristic Prediction Results

asset_id	pred_grade	actual_grade
327169	2	2.0
328183	2	2.0
328184	2	2.0
328185	2	2.0
328190	2	2.0
328191	2	2.0
102898	3	3.0
105361	3	3.0
126419	3	2.0
133876	3	3.0
134914	5	3.0

Predictions using asset-by-asset investigations are promising, with six out of six flood gates (assets with ids beginning 32… , Table 4.2) assessed predicted perfectly, and overall mean absolute error (MAE) was ~0.27 (Table 4.1). However, assessed embankments showed MAE of 0.6, with two of assets predicted to be worse than the actual grade. This is interesting for two reasons:

1. Direction of the error means these assets would be flagged as needing attention by a tool, potentially wasting resources
2. Embankments were the asset type requiring most manual work to assess, with multiple sources of imagery and manual elevation profiles drawn.

The main finding is confirmation that producing these assets on a national scale is infeasible, and therefore heuristics cannot be used accurately, with application of local knowledge and domain expertise, to construct a screening tool to help flexible scheduling of inspections because it would take too long to collect data. More importantly, all data collected is fixed, so the exact deterioration rate according to heuristics is known from the point of construction. No extra value can be extracted that would update information and mean that a flexible schedule could improve. This confirms the main advantage of a ML approach - new data can be collected to encode signal from events happening to flood defences during their lifetime or discovered after construction. The reason this investigation is impractical at a national scale is twofold: amount of time taken (several days) to compile trustworthy data for just eleven assets; and high rejection rate, with hundreds of assets screened out before all data necessary to select a curve was collected.

It is acknowledged selection of the correct curve would ordinarily be decided by an expert engineer with local knowledge, but since it was not possible to access this expertise, this investigation's findings support the project's premise that ML approaches hold promise.

4.2 National Scale Implementation

4.2.1 Rationale

In order to directly compare performance of ML models to heuristics and answer the research question, heuristic predictions must be calculated at comparible scale. Learnings from the Individual Heuristic Investigation confirm infeasibility of producing faithful predictions where each curve input is verified asset-by-asset. The best approach to benchmarking ML models becomes implementation using generalising assumptions to impute data unavailable in the AIMS register or other national datasets.

By doing so, the number of assets possible to create heuristic estimates for increased to 14,262, approximately 61.2% of assets.

4.2.2 Method

Data in the Halcrow document describing construction of heuristics was transcribed from tables into a python script to recreate deterioration curves. The best assumptions for assets having missing information were made, for example all embankments have a variable core material and narrow crest. Implementations of heuristics are found in notebook 8 for original investigation and `flood_defence_condition/src/heuristic.py` for final implementation. Once all inputs are confirmed, the correct curve can be selected and the condition grade is predicted using the asset age.

Selected heuristic curves are in Figure 4.2, full curves available in Figure C.2.

4.2.3 Results and Findings

The main results of heuristic prediction at scale will be presented in Chapter 7 and be evaluated in comparison with ML results and naive baselines. Table 4.1 contains MAE for immediate comparison with the full asset-by-asset investigation. This shows a large increase in error, demonstrating generalising assumptions don't hold, preventing heuristic application from being effectively scaled. If error of scaled predictions were in line with the first exercise, this would undermine the need for an ML approach. Instead, there are two key findings calling for a new approach to be developed: high error, almost as poor as naive predictors (shown in chapter 7); and lost coverage, with only 62% of assets used in ML experiments able to implement a curve, normally (~97% of missing) because asset age is not trustworth enough to use.

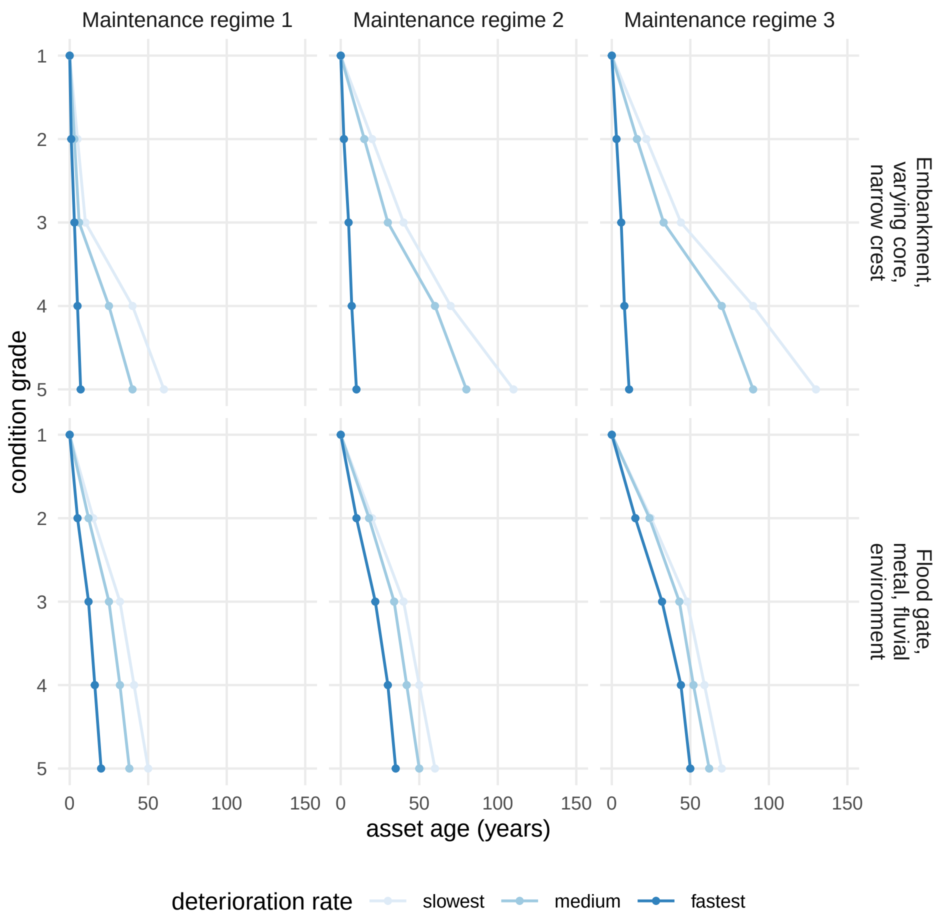

Figure 4.2: Halcrow deterioration curves for two asset families, by maintenance regime. Each line is one deterioration rate. Applying curves nationally requires selecting rate, which AIMS does not record.

Chapter 5

Ethical Consideration

All data collected and processed is used permissibly under their licences with all attributions given (appendix A1).

This project has low ethical risk because it does not concern human or animal wellbeing during research or development. The major consideration is avoiding misrepresentation of the artifact that could recommend deviating from current effective flood management. One source of bias necessary to raise is that data preparation systematically excludes assets from specific regions, often high population areas where failure of assets would matter most. For example, the assets in the Thames watershed tend to be excluded because they are redacted in the EA's EIR response due to third party management. Results may generalize poorly to these localities - it has not been tested so any use of findings must qualified accordingly. Another consideration is that EA information has been obtained and used with reproducibility and transparency in mind, while the EA itself seems committed to controlling disclosure of information, demonstrated through their redaction of assets deemed critical strategic infrastructure. The EA is tasked with preventing information disclosure weakening the national flood defence program, and must have seen potential for open information to increase risk. However, since all results hosted respect redactions, and only with open government licenced datasets, the public nature of the repository is judged not to have further ethic risk. Finally, the evaluation section will show modest results that, while answering the research question, do not meet the required bar for this artifact to be operationally useful. The evaluation must be honest in its assessment and recommendations.

Chapter 6

Methodology and Design

To meet the objective of being a reproducible research artifact, the code repository is designed for reuse and extension by future collaborators, whether researchers or practitioners. It contains source code as a locally installable package, designed to be managed with well known package manager uv, with configuration and lock files for identical installation across machines. Code is organised by into source code, experiments, and notebooks. Notebooks are designed as finished accounts of work completed, but are rerunnable once the project is set up. To allow for reproduction and extension of work, the repository contains a citation cff file and an MIT licence. Scripts used for one-off tasks are committed in their own folder.

All active research was carried out in this repository. Chronologically numbered notebooks contain EDAs for each data source, used to investigate feature extraction, engineering,and selection. Each experiment is a module with its own design, recording results and run logs within. Notebooks then interrogate results, and this is where findings are found and discussed.

The methodology of each stage of research is discussed in more depth:

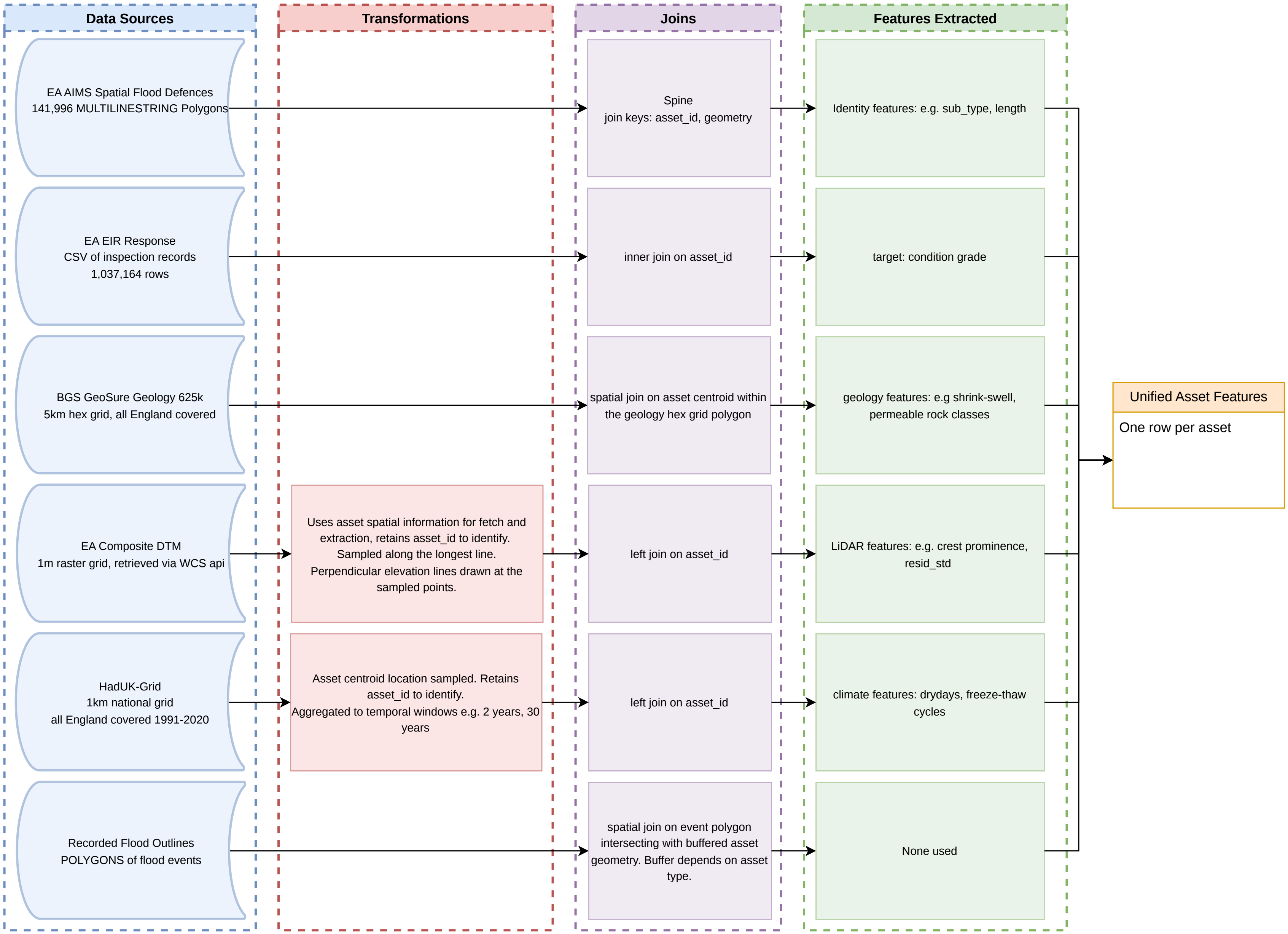

Figure 6.1: Data Collection and Integration

6.1 Experiments Design

These are loops that train and test many model variations, and emit results for each design plus benchmarks. The four experiments are:

1. Baseline: initial implementation of written src code with simplest models
2. Coursegrid: simple hyperparameter choices, main intent to test different feature selections
3. Fine grid: refined choice of features, granular choice of hyperparameters
4. Final grid: only two final feature compositions, fine mesh of hyperparameters known to contain optimum values

For each experiment, intent was known and evaluation was designed before it was run, meaning decisions about success and interesting findings stayed honest. Experiment design was nonetheless iterative: findings of each inform the next design, including the next feature compositions, choice of hyperparameter grids, and code architecture. Within each experiment, several ML-specific considerations were made:

6.1.1 Spatial Cross Validation

Flooding is a geographic phenomenon - it occurs beside rivers or coastlines, often on low-lying or flat areas. Human settlements and other valuable constructions follow other geographic rules. Its therefore unsurprising to see flood defences form geographic clusters. Plotting them makes it visually clear that condition grades are geographically correlated - e.g. a cluster of lower graded assets in the South East while the South West has mostly fair conditioned defences. This leads to a problem configuration where it could be possible to train a model that seems to succeed by making good predictions when the geographical location of the asset is encoded in one or more of the features. However, since it is desired that this project demonstrates it is possible to learn the condition of the asset from its own features, spatially blocked cross validation is used so metrics are not inflated when an asset nearby to training examples is evaluated, following recommendations in Roberts *et al.* (2017).

6.2 Model and Data choices

Table 6.1: Valid Asset Filter Funnel

Step	Rule	Remaining
EIR response	inspection records supplied under EIR	1,037,164 rows
Disclosed grade	drop 888 / 999 redaction sentinels and blanks	275,571 rows
One row per asset	most recent inspection retained	36,562 assets
Matched to AIMS	inner join on asset ID (141,996 assets in register)	35,270 assets
Interpretable grade	drop grade 0 (Unchecked)	35,202 assets
Engineered defence	Natural High Ground excluded as passive landform	23,304 assets

6.2.1 Ordinal Awareness

Condition grades are not five pure categories but an ordinal hierarchy - grade 1 is better condition than 2, which is better than 3 and so on. Evaluation must respect this. Models should be punished more for misclassifying a grade 2 asset as grade 5 than as grade 3. This means we would want metrics chosen to evaluate models to indicate a poorer performance by being lower (or higher, depending on metric). Wang *et al.* (2025) documents ordinal-aware classification methods, and describes the Frank and Hall (2001) ensemble method explored in this project.

6.2.2 Model Selection

Gradient boosting is a well known ML paradigm producing strong results, and handles null values and correlated features - important when working with messy data sets (James *et al.*, 2023). Histogram Gradient Boosting, a scikit-learn implementation of the LightGBM algorithm (scikit-learn developers, n.d.; Ke *et al.*, 2017) was chosen for its native handling of mixed categorical, continuous and discrete inputs, an advantage over ensembles like random forest. Boosting rather than bagging ensembles were chosen for better potential with unbalanced data, considering the main objective is to be good at predicting the minority classes (grades 4 and 5). Another advantage of tree-based ensembles is explainability. Given research investigates feasibility of developing an operational tool, ability to explain how decisions get made is valuable. Trees are a concept that can be grasped intuitively, and additionally allow explainability through feature importance and SHAP value analysis.

One way successful models were categorised was ability to classify well in the context of ordinal labelling, and to cope with significant class imbalance (~67% of all assets were grade 3). Class imbalance was so strong that a model simply predicting the median every time would achieve a fair MAE and accuracy, so these two metrics were relegated. Instead, quadratic weighted Cohen's kappa score (qwk) was identified (Bendavid, 2008). This rewards models performing well in ordinal-awareness, providing a scale that is sensible and interpretable in the context of class imbalance.

6.2.3 Other Considerations

Shapley additive explanations (SHAP) serve two purposes:

1. deciding which features impact models to inform feature selection
2. exploring which sources and types of data improve models, and at which grades, in order to understand how predictive signal is encoded in data sources (Muschalik *et al.*, 2024).

The experiment loops were designed knowing this evaluation was needed so that SHAP values were calculated per-model, per-fold, and emitted. Analysis is then done on individual cells or averaged values filtering to specific families of model configurations.

6.3 Evaluation Approach

A consistent target was defined as condition grade present in the EA's response to the Environmental Information Request (EIR), rather than the version in flux in the online AIMS portal. This was consistent and used for every model iteration, dummy classifiers, and heuristic classifiers to allow honest comparisons. These dummy classifiers were naive predictors that could be used as baselines. They were:

- a flat baseline always predicting the median (grade 3). This could do well on many metrics because of the severe class imbalance. However, it is known to have no skill so should perform poorly on any metrics that reward models for ability to perform well on minority classes and make ordinally aware predictions
- a classifier that draws the prediction randomly from the known distribution of condition grades. This allows this baseline to make some predictions at the interesting minority classes (4 and 5) but is still known to have no skill, so can be used as a the benchmark to beat in this part of the problem.

In early work, a wide variety of metrics were used, including but not limited to mean absolute error (MAE), precision, recall, and accuracy, and macro f1 score. We could see that MAE did not work for rewarding models with better ordinal awareness so QWK was adopted early in the project. For final experiments, area under the precision-recall curve considering predictions for grade 4 and 5 only (PR-AUC ≥ 4) was used as a headline statistic focused on rewarding models that performed well and indicated they had learned skill at predicting the operationally significant classes, as well as qwk for ordinal awareness, and f1 score for grades as a supplementary metric. Reliance on F1 score was decreased because it depends on the probability threshold being set first. F1 score doesn't discriminate predictions whether they have 0.65 probability of being class 4 or 0.95. Once it became clear that the artifact would have to set the threshold high or use the probability to rank the most confident predictions, F1 became hard to use on the whole dataset while PR-AUC was still useful.

Overall evaluation of the artifact will include:

- performance of models, directly relating to the research question

- how the artifact has uncovered information necessary to make good predictions
- whether the artifact helps inform future work through findings and reproducibility.

Chapter 7

Results

7.1 High Level

ML models outperform naive baselines and heuristic predictions at the important condition ≥ 4 boundary when a tightly scoped budget of the few hundred most confident predictions is considered, on all important metrics inluding PR-AUC and QWK. These are shown in Table 7.1 and Table 7.2.

Performance degrades as lower confidence predictions are included eventually falling back to the no-skill baseline.

Several features from additional remote sensed or environmental datasets were used in the final model because they carried predictive signal, but the majority of features engineered in this project were discarded for having little or no potential to improve predictions.

The PR-AUC at $grade \geq 4$ results showed no evidence heuristic predictions applied at national scale (as described in Chapter 4.2) had more skill than naive predictors - scores are almost identical. However, QWK is significantly higher, indicating predictions made by heuristic follow the ordinal rules of the classes.

7.2 Evidence of Predictive Signal

Figure 7.1 from the coarse grid experiment comparing several families of feature selections shows that predictive signal exists among extracted features. This precision curve against assets flagged (from most to least confident prediction, using the predict_proba value) was chosen because recall is fairly consistent across asset budgets sizes so precision is the main discriminating metric. The curve covers only classifications of assets grade 4 or 5 vs lower - the boundary between fair condition and poor condition, and therefore the point of most concern. Every feature selection in this experiment showed an improvement over the "no skill" base rate once the threshold is set appropriately, and over identity features alone at some threshold. These results informed the more refined feature selection designs for final experiments. Most additional features carried only weak signal, and including degrade models due to the curse of dimensionality faster than any improvement from additional information encoded was gained. Figure 7.3 shows final configs selected and the precison curve from the final experiment. Tables Table C.1 in the appendix describe exact feature selection. This indicates better ML models are achievable with extra environmental, remote sensed, or climate data augmenting data held in the AIMS national flood defense register. These features are different from the static fields used in Halcrow heuristics, and add skill to the classifier. Figure 7.2 shows LiDAR as the group of features improving model performance most. Figure 7.3 shows heuristic predictions sit at the same base rate as naive predictors, so while they are usually sensible (QWK=1.85, comfortably above dummies), they have no discrimination in the operationally significant classes (PR-AUC=0.099) while ML models do.

Each source has its own notebook in the repository where far more features were investigated. The flood event exposure data source (whether an asset has been exposed to a flood thats happened in a defined time window) was completely unused because none of its features were useful.

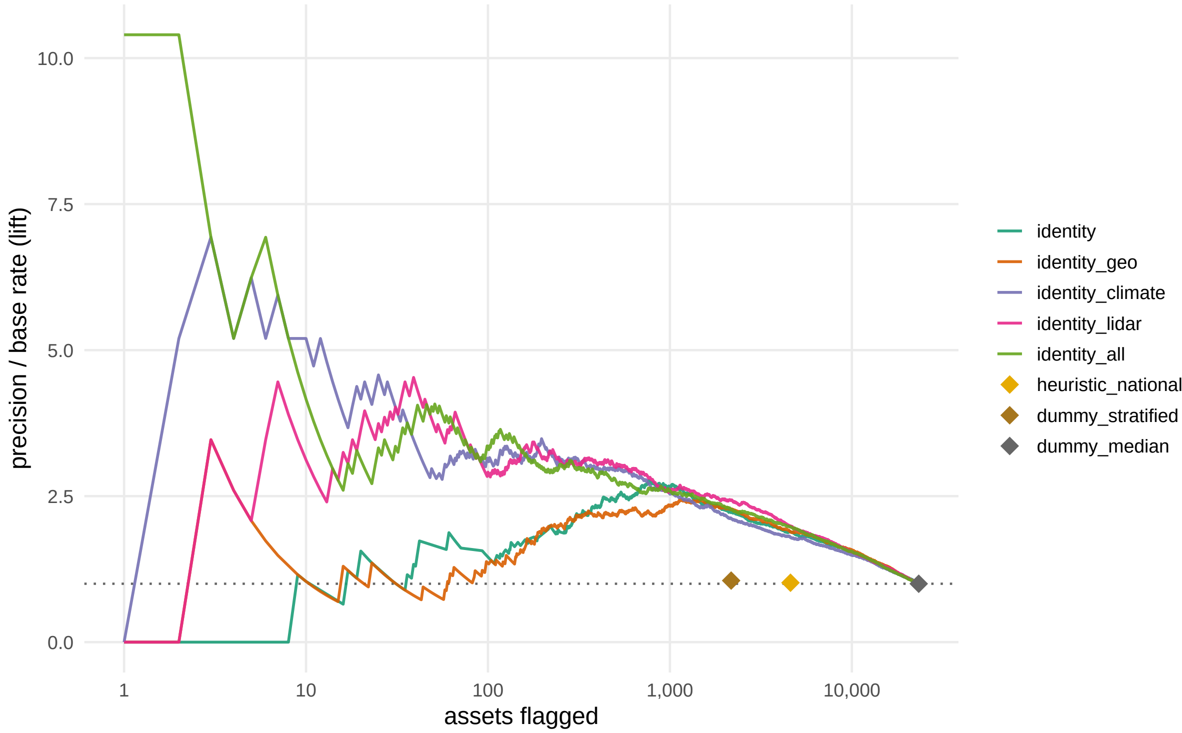

Figure 7.1: Precision improvement against inspection scope for the best cell per feature family, coarse grid experiment. Baselines shown at maximum lift operating point. Base rate 0.096.

The curves also show that precision collapses back down to the base rate as more and more assets are added into the pool. Once all predictions are used (no confidence threshold set), the ML models are no more useful than the naive predictors.

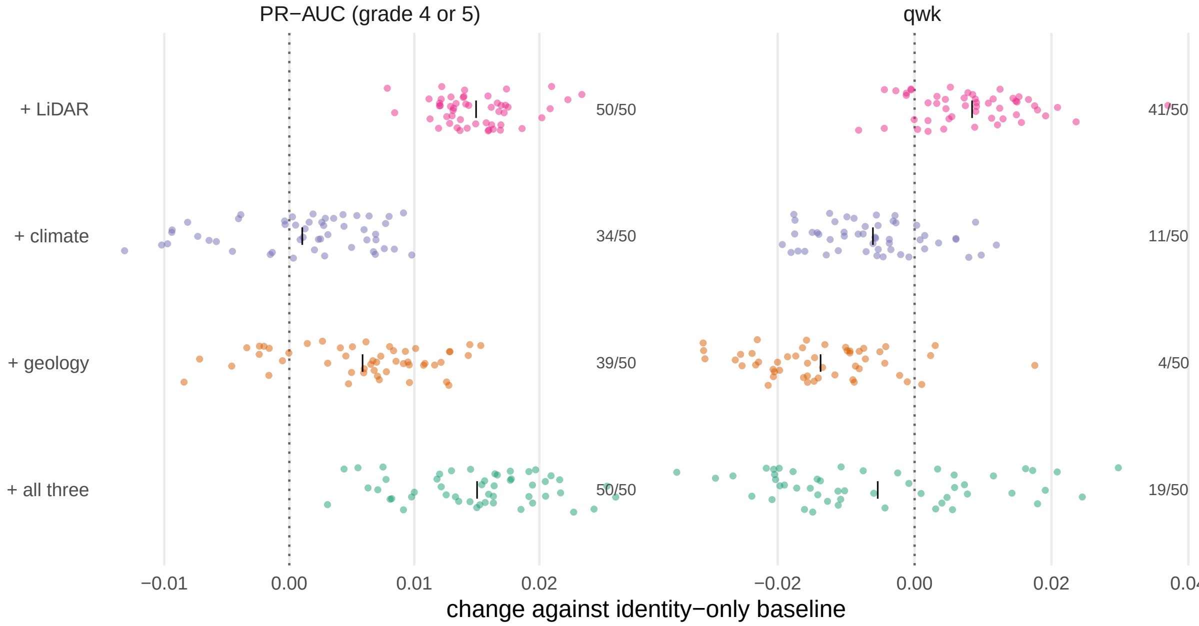

Figure 7.2: Effect of adding each feature group to the baseline (only identity features). Data from the coarse grid experiment. Each point is one hyperparameter config, paired within fold with the same configuration on just identity features. Bars mark the mean across the 50 configurations, and the count gives configurations with a positive delta.

Table 7.1: Confusion comparison of two HGB models and the Halcrow heuristics

classifier	True Pos	False Pos	False Neg	True Neg	Precision	Recall	Improvement
HGB, identity features + LiDAR + climate	67	133	2,174	20,930	33.5%	3.0%	3.48×
HGB, identity + LiDAR + all	60	140	2,181	20,923	30.0%	2.7%	3.12×
Halcrow heuristic	24	176	2,217	20,887	12.0%	1.1%	1.25×

Table 7.2: Argsweep results: Performance of classifiers on all 23,304 assets

classifier	Feature Composition	pr auc @ grade >= 4	Quadratic Weighted Cohen's Kappa
HGB	Identity + Lidar + Climate	0.19765	0.280176
HGB	Identity + All Lidar	0.19595	0.28451
Nationally Applied Heuristics	-	0.09884	0.18514
Naive dummy - stratified	-	0.09751	-0.00085
Naive dummy - median	-	0.09658	0.00000

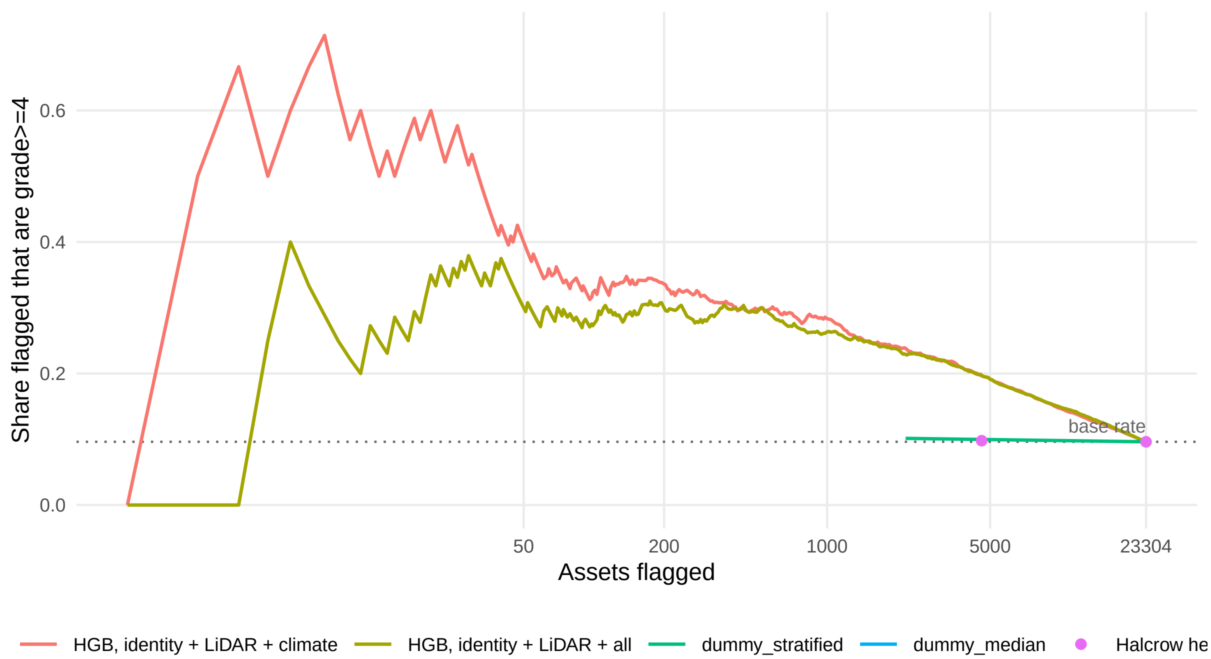

Figure 7.3: Precision against inspection budget. The heuristic is drawn at its available operating points, with constant score dummies omitted.

7.3 SHAP Values

SHAP values were inspected to learn each feature's importance and contibution to classification decisions. Charts like Figure 7.4 were used consistently within experiments in notebooks in the repository to inform feature selection. Figure 7.4 shows the best performing feature composition, with SHAP values aggregated across a few similar hyperparameter configurations. The bump chart was chosen because it shows some features become important at the operationally important grade 3|4 boundary, while others fall off or stay consistent. For example, `aims__asset_sub_type` is the most important feature for making correct decisions when a fair grade (1-3) is predicted, but drops to become the second least important feature for predicting grade 5. Conversely, the `lidar__crest_resid_std` feature is second least used when classifying grade 3 but jumps to second most used for grade 4 on the other side of the boundary. Age is consistently an important feature. This chart does not show magnitude, only rank - read outs of exact scores and beehive plots in the experiment notebooks show that top features tend to be much stronger and the final five or so features are very close in absolute value.

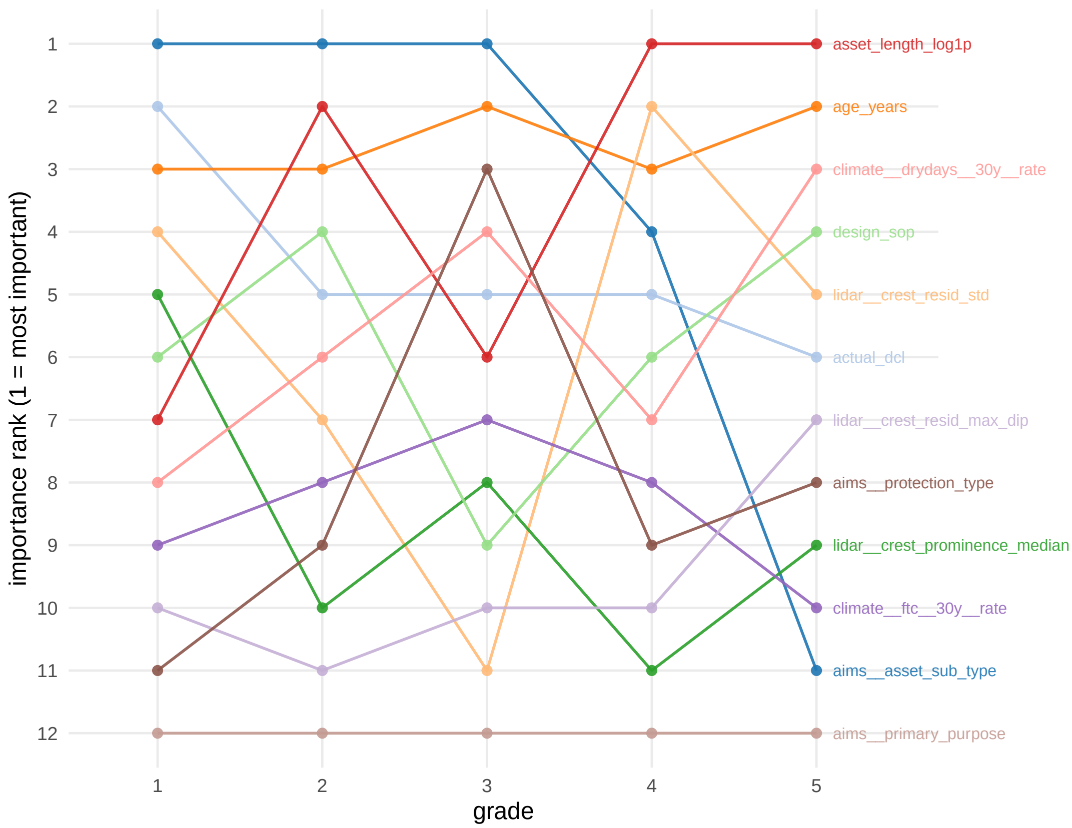

Figure 7.4: SHAP feature importance rank across condition grades, HGB with identity, LiDAR and raw climate features. Rank 1 is the most important feature within a grade.

Chapter 8

Evaluation

8.1 Evaluation of Artifact

ML classifiers produced modest results and are not ready for an operationally useful flood management tool. They did outperform baselines and heuristics, answering the research question by suggesting ML could be advantageous for developing alternative flood management tools. However, if performance represents a ceiling, future work should look for alternatives. The most likely explanation for modest performance is lack of predictive power in the data. Future work could: identify data sources with stronger signal, explore collection and use of directly sensed data like local hardware at defence locations, or work with infrastructure engineers to identify other ways to capture and encode domain expertise.

Findings on which features carried predictive power are interesting but possibly incomplete. Far more time is needed to search for sources and dedicate necessary depth to feature extraction formulations for each. For example many ideas for extracting damage encoding features from LiDAR profiles were planned then rejected because of time constraints. That said, findings stand alone as genuine contributions describing relative predictive signal in several previously unconnected data sources.

The findings presented are considered trustworthy given robust experimental design: consideration of spatial leakage, controlled evaluation on shared cross-validation folds, careful progression through tuning grids, and care over data collection and preparation.

8.2 Evaluation of General Approach

The data scope should have been narrower given time and resources, although this would likely lead to a less successful artifact.

Supervision sessions were of great help, for technical discussion, developing narrative structure, and above all for constantly evaluating and realigning the scope. Feedback was acted on; discussion points were normally recorded next day. (Table B.1) Milestones were agreed in the first session and updates provided as they were achieved. On reflection, scheduling sessions regularly and far in advance would have helped consistency by creating regular checkpoints.

Research was done iteratively; new literature material was sought, read, and reread throughout. This approach was valuable and made the project a rich learning experience, although tighter project-management-style phases could have delivered a more polished tool with more obvious practical value, at the cost of pure research output and domain learning. Balance had to be found.

Chapter 9

Conclusion

9.1 Project Summary

This project researched the EA's challenge of maintaining a large, aging flood defence infrastructure with a demanding inspection burden. Established engineering knowledge was reviewed, then a new dataset built from open sources to train ML models, attempting to outperform heuristic-derived predictions. Results of ML research were evaluated.

9.2 Results Summary

ML models produced outperformed heuristic-derived methods in the section of the dataset most relevant to developing a tool for practitioners, with precision ~3.5x the base rate versus 1.25x for heuristics. The research also reviewed a non-expert's application of the heuristics and produced findings on which features and data sources improve these classifiers. Section 2.1 set two bars: how well outputs answer the research question; and to what extent the artifact demonstrates operational usefulness. Following discussions in chapter 8, the research bar is considered met, although not the operational bar.

9.3 Discussion

9.3.1 Have aims been met?

The project collected several new data sources, and explored how ML models trained on these performed relative to heuristics. The was done reproducibly, with the repository publicly available for reproduction and extension. Findings about which asset characteristics and environmental features contribute most to predicting defence condition were presented. This is all evidence the aims were met. However, evidence that an operational tool could be developed following this methodology is weak.

9.3.2 Is the problem solved?

Good progress was made in the problem space, with new insight into applying Halcrow guidance, and successful feature engineering approaches and model architectures explored. Although knowledge has been contributed, findings are not strong enough to immediately improve how England's flood defences are managed.

9.4 Future Work

Future work is broad and abundant. Promising directions include using approaches from LiDAR feature extraction for ML training to revisit Halcrow curves, using environmental and physical features to develop updated, more informed heuristics. Additionally, ML models could be revisited for the same problem with new data, either new feature formulations from sources already explored, such as sensitivity testing on buffer zones to past flood event exposure, or new data entirely. Of particular interest would be data that could be captured from local or on-site stations, such as sensing small earth movements or real-time water velocities or earth temperatures. An extremely helpful data source possible with EA disclosure would be a schedule of maintenance and repair carried out on assets.

References

Abib Ali Awan (2023) *An Introduction to SHAP Values and Machine Learning Interpretability* [online]. Available from: https://www.datacamp.com/tutorial/introduction-to-shap-values-machine-learning-interpretability [Accessed 17 May 2025].

Ali, A.J. and Ahmed, A.A. (2025) Aquifer-specific flood forecasting using machine learning: A comparative analysis for three distinct sedimentary aquifers. *Science of The Total Environment* [online]. 1004, p. 180756. Available from: https://www.sciencedirect.com/science/article/pii/S0048969725023964 [Accessed 10 February 2026].

Anon. (no date a). *The Southern Coastal Group and SCOPAC* [online]. Available from: https://southerncoastalgroup-scopac.org.uk/capital-investment-programme/ [Accessed 17 February 2026].

Anon. (no date b). *Water Projects: Case Studies* [online]. Available from: https://waterprojectsonline.com/case-studies/ [Accessed 12 July 2026].

Anon. (no date c). *GOV.UK* [online]. Available from: https://www.gov.uk/government/publications/flood-risk-asset-maintenance-and-inspection-good-practice-guidance [Accessed 22 February 2026].

Anon. (no date d). *API Catalogue* [online]. Available from: https://www.api.gov.uk/ea/hydrology/ [Accessed 17 February 2026].

Anon. (no date e) [online]. Available from: https://www.nationalarchives.gov.uk/doc/open-government-licence/version/3/ [Accessed 27 August 2026].

Bates, P.D., Savage, J., Wing, O., Quinn, N., Sampson, C., Neal, J. and Smith, A. (2023) A climate-conditioned catastrophe risk model for UK flooding. *Natural Hazards and Earth System Sciences* [online]. 23 (2), pp. 891–908. Available from: https://nhess.copernicus.org/articles/23/891/2023/ [Accessed 10 February 2026].

Beltrán, A., Maddison, D. and Elliott, R. (2019) The impact of flooding on property prices: A repeat-sales approach. *Journal of Environmental Economics and Management* [online]. 95, pp. 62–86. Available from: https://linkinghub.elsevier.com/retrieve/pii/S0095069617303066 [Accessed 10 February 2026].

Beltrán, A., Maddison, D. and Elliott, R.J.R. (2018) Assessing the Economic Benefits of Flood Defenses: A Repeat-Sales Approach. *Risk Analysis* [online]. 38 (11), pp. 2340–2367. Available from: https://onlinelibrary.wiley.com/doi/10.1111/risa.13136 [Accessed 17 February 2026].

Bendavid, A. (2008) Comparison of classification accuracy using Cohen's Weighted Kappa. *Expert Systems with Applications* [online]. 34 (2), pp. 825–832. Available from: https://linkinghub.elsevier.com/retrieve/pii/S0957417406003435 [Accessed 30 June 2026].

Bonnier, T. and Bosch, B. (2022) Assessing the Robustness of Ordinal Classifiers against Imbalanced and Shifting Distributions. *Proceedings of the Fourth International Workshop on Learning with Imbalanced Domains: Theory and Applications* [online]. PMLR, pp. 112–126. Available from: https://proceedings.mlr.press/v183/bonnier22a.

html [Accessed 17 February 2026].

Brown, C., Chatterton, J. and Purcell, A. (2014) *Asset performance tools: Asset inspection guidance* [online]. Environment Agency [SC110008/R2]. Available from: https://www.gov.uk/flood-and-coastal-erosion-risk-management-research-reports/flood-risk-asset-inspection-research-to-improve-interventions [Accessed 22 February 2026].

Cesare, M., Santamarina, J., Turkstra, C. and Vanmarcke, E. (1992) Modeling Bridge Deterioration with Markov Chains. *Journal of Transportation Engineering-asce - J TRANSP ENG-ASCE* [online]. 118.

Chen, T. and Guestrin, C. (2016) XGBoost: A Scalable Tree Boosting System. *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining* [online]. pp. 785–794. Available from: http://arxiv.org/abs/1603.02754 [Accessed 29 April 2025].

Clarke, J.S. (2023) *Thousands of England's flood defences were in poor condition before storms hit. Unearthed* [online]. Available from: https://unearthed.greenpeace.org/2023/10/30/flood-defences-storm-babet-england/ [Accessed 10 February 2026].

Coxon, G. *et al.* (2020) *Catchment attributes and hydro-meteorological timeseries for 671 catchments across Great Britain (CAMELS-GB)* [online]. Available from: https://doi.org/10.5285/8344e4f3-d2ea-44f5-8afa-86d2987543a9.

Coxon, G. *et al.* (2025) *Catchment boundaries, daily and sub-daily hydrometeorological time series, groundwater level time series and attributes for 671 catchments in Great Britain (CAMELS-GB V2)* [online]. Available from: https://doi.org/10.5285/9a46d428-958f-4ac1-86eb-94eee70c0955.

Dawson, G., Butt, J., Jones, A. and Fraccaro, P. (2023) Flood susceptibility mapping at the country scale using machine learning approaches. *Applied AI Letters* [online]. 4 (4), p. e88. Available from: https://onlinelibrary.wiley.com/doi/abs/10.1002/ail2.88 [Accessed 10 February 2026].

Environment Agency (2025) *AIMS Defence (Spatial Flood Defences) Product Description* [online]. Bristol. Available from: https://environment.data.gov.uk/dataset/8e5be50f-d465-11e4-ba9a-f0def148f590.

Environment Agency (2020) *AIMS Spatial Flood Defences (inc. Standardised attributes)* [online]. Available from: https://environment.data.gov.uk/dataset/8e5be50f-d465-11e4-ba9a-f0def148f590 [Accessed 17 February 2026].

Environment Agency (no date a) *Defra Data Services Platform* [online]. Available from: https://environment.data.gov.uk/dataset/13787b9a-26a4-4775-8523-806d13af58fc [Accessed 27 August 2026].

Environment Agency (2021) *Flood and Coastal Erosion Risk Management: Capital Investment Programme (2021-2027)* [online]. Available from: https://environment.data.gov.uk/asset-management/downloads/capital-programme.pdf [Accessed 17 February 2026].

Environment Agency (no date b) *Item: /asset-management/doc/reference* [online]. Available from: https://environment.data.gov.uk/asset-management/doc/reference [Accessed 17 February 2026].

Environment Agency (2022) *Partnership funding calculator 2020 for FCERM grant-in-aid (GIA). GOV.UK* [online]. Available from: https://www.gov.uk/government/publications/partnership-funding-calculator-2020-for-fcerm-grant-in-aid-gia [Accessed 17 February 2026].

Environment Agency (2026) *Programme of flood and coastal erosion risk management (FCERM) schemes. GOV.UK* [online]. Available from: https://www.gov.uk/government/publications/programme-of-flood-and-coastal-erosion-risk-management-schemes [Accessed 12 July 2026].

Environment Agency (no date c) *Rainfall API reference* [online]. Available from: https://environment.data.gov.

uk/flood-monitoring/doc/rainfall [Accessed 17 February 2026].

Fard, F. and Fard, F.S.N. (2024) Development and Utilization of Bridge Data of the United States for Predicting Deck Condition Rating Using Random Forest, XGBoost, and Artificial Neural Network. *Remote Sensing* [online]. 16 (2). Available from: https://www.mdpi.com/2072-4292/16/2/367 [Accessed 10 February 2026].

Feng, K., Lin, N., Kopp, R.E., Xian, S. and Oppenheimer, M. (2025) Reinforcement learning–based adaptive strategies for climate change adaptation: An application for coastal flood risk management. *Proceedings of the National Academy of Sciences* [online]. 122 (12), p. e2402826122. Available from: https://www.pnas.org/doi/10.1073/pnas.2402826122 [Accessed 10 February 2026].

Ficarra, M. and Mari, R. (2025) *Weathering the storm: Sectoral economic and inflationary effects of floods and the role of adaptation.* Staff {{Working Paper}}. Bank of England [1120].

Frank, E. and Hall, M. (2001) A Simple Approach to Ordinal Classification. In: Goos, G., Hartmanis, J., Van Leeuwen, J., De Raedt, L. and Flach, P., eds. *Machine Learning: ECML 2001* [online]. 2167, Berlin, Heidelberg: Springer Berlin Heidelberg, pp. 145–156. [Accessed 26 August 2026].

Géron, A. (2023) *Hands-on machine learning with Scikit-Learn, Keras and TensorFlow: Concepts, tools, and techniques to build intelligent systems.* Third edition. Beijing: O'Reilly Media, Inc.

Gideon Amos (2026a) *Written questions and answers - Coastal Erosion and Flood Control. UK Parliament* [online]. Available from: https://questions-statements.parliament.uk/written-questions/detail/2026-02-11/112920/ [Accessed 25 August 2026].

Gideon Amos (2026b) *Written questions and answers - Flood Control. UK Parliament* [online]. Available from: https://questions-statements.parliament.uk/written-questions/detail/2026-02-11/112916/ [Accessed 25 August 2026].

Gideon Amos (2026c) *Written questions and answers - Flood Control: Finance. UK Parliament* [online]. Available from: https://questions-statements.parliament.uk/written-questions/detail/2026-02-11/112921/ [Accessed 25 August 2026].

Grus, J. (2019) *Data Science from Scratch: First Principles with Python.* 2nd ed. Sebastopol: O'Reilly Media, Incorporated.

Halcrow Group Limited (2013) *Delivering benefits through evidence: Practical guidance on determining asset deterioration and the use of condition grade deterioration curves:Revision 1Report - SC060078/R1.* Bristol: Environment Agency.

Hannaford, J. *et al.* (2022) *Hydrological projections for the UK, based on UK climate projections 2018 (UKCP18) data, from the enhanced future flows and groundwater (eFLaG) project* [online]. Available from: https://doi.org/10.5285/1bb90673-ad37-4679-90b9-0126109639a9.

Hao, A., Zhu, C., Zhu, J., Wu, J. and Chu, X. (2026) *Ranking-aware Reinforcement Learning for Ordinal Ranking* [online]. Available from: http://arxiv.org/abs/2601.20585 [Accessed 17 February 2026].

Hill, D.M. (no date) *Flood_defence_condition* [online]. Available from: https://github.com/danielmh111/flood/_defence/_condition/#.

HM Government (2026) *National Risk Register 2026* [online]. Available from: https://assets.publishing.service.gov.uk/media/6a50d66b1228eb26a4cab76c/National/_Risk/_Register/_2026.pdf [Accessed 26 August 2026].

Hollis, D., McCarthy, M., Kendon, M., Legg, T. and Simpson, I. (2019) HadUK-Grid—A new UK dataset of gridded climate observations. *Geoscience Data Journal* [online]. 6 (2), pp. 151–159. Available from: https:

//rmets.onlinelibrary.wiley.com/doi/10.1002/gdj3.78 [Accessed 17 February 2026].

James, G., Witten, D., Hastie, T., Tibshirani, R. and Taylor, J.E. (2023) 'An introduction to statistical learning: With applications in Python'Springer texts in statistics. Cham: Springer.

Jones, A. *et al.* (2023) AI for climate impacts: Applications in flood risk. *npj Climate and Atmospheric Science* [online]. 6 (1), p. 63. Available from: https://www.nature.com/articles/s41612-023-00388-1 [Accessed 10 February 2026].

Ke, G., Meng, Q., Finley, T., Wang, T., Chen, W., Ma, W., Ye, Q. and Liu, T.-Y. (2017) LightGBM: A Highly Efficient Gradient Boosting Decision Tree [online]. Available from: https://www.researchgate.net/publication/378480234/_LightGBM/_A/_Highly/_Efficient/_Gradient/_Boosting/_Decision/_Tree [Accessed 1 September 2026].

Kobayashi, K., Kaito, K. and Lethanh, N. (2012) A statistical deterioration forecasting method using hidden Markov model for infrastructure management. *Transportation Research Part B: Methodological* [online]. 46 (4), pp. 544–561. Available from: https://linkinghub.elsevier.com/retrieve/pii/S0191261511001809 [Accessed 9 February 2026].

Matias de Oliveira, J.L. (no date) *A Heterogeneous Markov Chain Model to Predict Pavement Deterioration and Optimize Repair Activities*. PhD thesis. University of Minnesota.

Mia, M.M. and Kameshwar, S. (2023) Machine learning approach for predicting bridge components' condition ratings. *ResearchGate* [online]. Available from: https://www.researchgate.net/publication/374552913/_Machine/_learning/_approach/_for/_predicting/_bridge/_components'/_condition/_ratings [Accessed 10 February 2026].

Muschalik, M., Baniecki, H., Fumagalli, F., Kolpaczki, P., Hammer, B. and Hüllermeier, E. (2024) *Shapiq: Shapley Interactions for Machine Learning* [online]. Available from: https://arxiv.org/abs/2410.01649 [Accessed 18 August 2026].

Nasiri Khiavi, A., Vafakhah, M., Kim, D., Jun, C. and Bateni, S. (2025) Integrating Remote Sensing, Machine Learning, and Local Knowledge for Innovative Flood Susceptibility and Vulnerability Mapping. *Journal of Flood Risk Management* [online]. 18.

National Audit Office (2020) *Managing flood risk*. London: National Audit Office [HC 962].

National Audit Office (2023) *Resilience to flooding*.

Rakesh Choudhary, Ajay Kumar, Priya Dharsini and Mude Murali Naik (2025) Predicting water quality index using stacked ensemble regression and SHAP based explainable artificial intelligence. *ResearchGate* [online]. Available from: https://www.researchgate.net/publication/394890982/_Predicting/_water/_quality/_index/_using/_stacked/_ensemble/_regression/_and/_SHAP/_based/_explainable/_artificial/_intelligence [Accessed 22 February 2026].

Roberts, D.R. *et al.* (2017) Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure. *Ecography* [online]. 40 (8), pp. 913–929. Available from: https://nsojournals.onlinelibrary.wiley.com/doi/10.1111/ecog.02881 [Accessed 18 August 2026].

Rondinone, M., Dal Sasso, S.F., Aung, H., Contillo, L., Dimola, G., Schiattarella, M., Fiorentino, M. and Telesca, V. (2025) Assessing Flood and Landslide Susceptibility Using XGBoost: Case Study of the Basento River in Southern Italy. *Applied Sciences* [online]. 15, p. 5290.

Ruth Power (no date) *In Defence of Flood Defence Maintenance | ABI* [online]. Available from: https://www.abi.org.uk/news/blog-articles/2021/10/in-defence-of-flood-defence-maintenance/ [Accessed 10 February 2026].

scikit-learn developers (no date) *HistGradientBoostingClassifier*. *scikit-learn* [online]. Available from: https://scikit-learn/stable/modules/generated/sklearn.ensemble.HistGradientBoostingClassifier.html [Accessed 1 September 2026].

Skouralis, A. and Lux, N. (2023) *The impact of flood risk on England's property market.* London: Bayes Business School.

Skouralis, A., Lux, N. and Andrew, M. (2024) Does flood risk affect property prices? Evidence from a property-level flood score. *Journal of Housing Economics* [online]. 66, p. 102027. Available from: https://www.sciencedirect.com/science/article/pii/S1051137724000469 [Accessed 17 February 2026].

Tarrant, O., Hambidge, C., Hollingsworth, C., Normandale, D. and Burdett, S. (2017) Identifying the signs of weakness, deterioration and damage to flood defence infrastructure from remotely sensed data and mapped information. *Journal of Flood Risk Management* [online]. 11.

UK Centre for Ecology & Hydrology (no date a) *Hydrology API / Access to live data* [online]. Available from: https://hydrosensordata.ceh.ac.uk/help?gad/_source=1/&gad/_campaignid=21257924601/&gbraid=0AAAAA9T/_B6s9ePF9rA3jyw1Bdm6bPf8xp/&gclid=CjwKCAiAwNDMBhBfEiwAd7ti1ET7lX8JKz4UzE6mHZEHa8nLIjH_BwE [Accessed 17 February 2026].

UK Centre for Ecology & Hydrology (no date b) *National River Flow Archive* [online]. Available from: https://nrfa.ceh.ac.uk/ [Accessed 21 February 2026].

Wang, J., Chen, J., Liu, J. and Tang, D. (2025) *A Survey on Ordinal Regression: Applications, Advances, and Prospects* [online]. Available from: https://arxiv.org/html/2503.00952v1 [Accessed 17 February 2026].

Appendix A

Attributions

A.1 Data usage

The project makes use of data available under Open Govenment Licences:

- The AIMS flood defence data set, © Environment Agency copyright and/or database right 2020. All rights reserved.
- Contains British Geological Survey materials ©NERC 2026"
- LIDAR DTM composite 1m © Environment Agency copyright and/or database right 2022. All rights reserved.
- Recorded Flood Outlines © Environment Agency copyright and/or database right 2025. All rights reserved.
- Climate data obtained via CEDA (MET Office 2018)

Met Office; Hollis, D.; McCarthy, M.; Kendon, M.; Legg, T.; Simpson, I. (2018): HadUK-Grid gridded and regional average climate observations for the UK. Centre for Environmental Data Analysis, 21/082026. http://catalogue.ceda.ac.uk/uuid/4dc8450d889a491ebb20e724debe2dfb

A.2 AI Usage

Generative AI has been used at times in this project, in compliance with the recommended use of AI for this module. Models from Anthropic including Claude Sonnet 4.6, Claude Sonnet 5, Claude Haiku 4.5, Claude Opus 4.8, and Claude Opus 5 were used. Tasks that AI was used for include:

- Code debugging
- Limited use for code writing, for example the final experiment code based on the previous ones.
- Technical troubleshooting, such as setting up a r virtual environment and dvc setup
- Review of early notes and notebook markdown sections to get feedback on clarity and accuracy
- Transcribing python to r visualisations

All outputs were read and reviewed carefully before being included in the project.

AI was never used:

- to write any of the text in the body of this report
- to undertake experimental design
- to interpret results

Appendix B

Record of Supervisons

Table B.1: Record of supervisions and discussions

Date	Topics discussed	Actions agreed/taken
2026-09-01	Report structure and formatting; niche referencing best practices; application of marking rubric to chapter sections; licencing considerations.	
2026-07-21	progress through technical part of the project and realistic time boxes for the write up; final report structure, specifically what part of my currently produced research will fit into which chapters; some of the current findings and which will become findings worth writing up; briefly touched on ordinal awareness and why certain metrics are appropriate	
2026-07-14	Discussion of appropriate metrics, including why MAE wasn't appropriate, investigation of Cohen's Kappa and quadratic weighted Cohen's Kappa, why pr-auc is useful, why focusing on metrics at grade 4 and 5 is valuable	Use qwk and pr-auc at grade >= 4 as headline stats to guide next experiments
2026-07-07	Update of progress; explaination of heuristic limitations found; discussion on how to construct a narrative throughline; discussion of class imbalance challenge and how it will effect model performance and choice of evaluation metrics	Decided on items to discuss in the next meeting: Scope of producing a limited number of perfect heuristic predictions; the current performance of ML models and what good/adequate performance means in context; what metrics are appropriate
2026-06-30	Meeting cancelled, progress update shared by email and some of Kat's questions on ml approaches answered by explaining categorical and discrete data constraints	

Date	Topics discussed	Actions agreed/taken
2026-05-26	Ethics Considerations; Division between minimum viable project goals and stretch goals; In depth discussion of what constitutes an artifact and how artifact quality is weighed against holistic research output when marking the project; discussion of mathematical and visualisation techniques for modelling and quantifying state changes	Ethics form completed and sent to Kat; prioritise finished data processing so that I can be more confident when discussing which techniques will and won't work

Appendix C

Data Dictionary

Table C.1: Data dictionary describing features used in modelling

feature	source	type	definition
aims___asset_sub_type	EA AIMS	categorical	Recorded asset type (e.g. embankment, wall, flood gate, demountable defence, high ground and related classes) according to the AIMS flood defence register. Used in the model as an ”identity” feature.
aims___primary_purpose	EA AIMS	categorical	Primary function the asset was built to serve, according to the AIMS Flood defence register. Used in models as an ”identity” feature.
aims___protection_type	EA AIMS	categorical	Class of flood source the asset defends against (e.g. fluvial, tidal, coastal). Used in models as an ”identity” feature.
asset_length_log1p	EA AIMS	continuous	Natural log of asset centreline length in metres + 1. The asset centreline comes from the AIMS Flood Defence register. Log transform applied to correct the heavy right skew of raw length. Used in models as an ”identity” feature.
age_years	EA AIMS + EIR	continuous	Years between asset_start_date and the date of the inspection that reported the condition grade. Null where asset_start_date is absent - a subset of recorded start dates follow boilerplate patterns and are treated as unverified.
actual_dcl	EA AIMS	continuous	Recorded actual defence crest level, metres above ordnance datum (mAOD). Correlates with design_dcl at rho > 0.9998, so only one is used in modelling.
design_sop	EA AIMS	continuous	Design standard of protection, expressed as the return period in years of the event the asset was designed to withstand.
lidar___crest_resid_std	EA LIDAR Composite DTM 1m	continuous	Standard deviation in metres of the crest elevation residual (calculated along the length of the crest), after removing the elevation trend of the surrounding ground from the crest profile. This is designed to measure any uneven settling or movement in the asset, removing signal like absolute height and difference in height on each side. Has to be interpretted with the survey noise of +/- 0.15m.

Table C.1: Data dictionary describing features used in modelling *(continued)*

feature	source	type	definition
lidar___crest__resid__max_di	EA LIDAR Composite DTM 1m	continuous	Largest single downward dip in metres of the crest of the asset along its profile, after removing any trend such as a continuous slope. Its the deepest local low point relative to the local trend.
lidar___crest__prominence__r	EA LIDAR Composite DTM 1m	continuous	Median across sampling points of the prominence in metres of the crest above the surrounding cross section. Cross section means sampling across the asset, perpendicular to the crest, so this represents how prominant the asset is relative to the surrounding terrain on either side.
climate___drydays___30y___	HadUK-Grid (Met Office)	continuous	Count of dry days per observable year over the 30 year window ending at the inspection date, sampled at the asset centroid. Rate rather than count to remove signal that could be shared with the asset age.
climate___ftc___30y___rate	HadUK-Grid (Met Office)	continuous	Count of freeze-thaw cycle days per observable year over the 30 year window ending at the inspection date, sampled at the asset centroid. Air temperature is a proxy for the material temperature

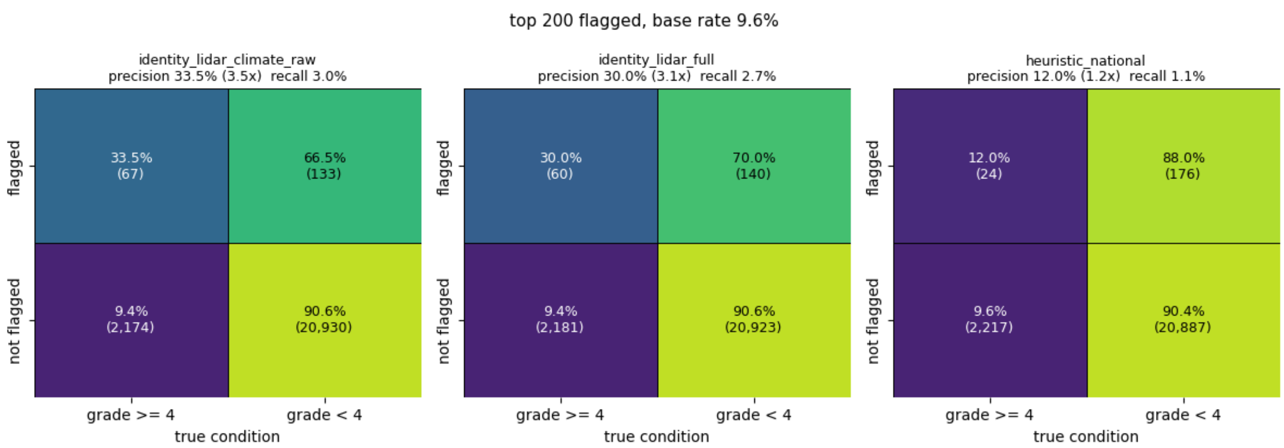

Figure C.1: Final Confusion Matrices

Table C.2: Hyperparameters of the reported models. Metrics are means across the ten cv folds.

Parameter	Selected	Best PR-AUC (climate)	Best PR-AUC (full)	Best QWK
`cell_id`	`4e993f867560`	`8de98379fe6f`	`57eee29de136`	`4009966c686`
Composition	`identity_lidar_climate_`	`identity_lidar_climate_`	`identity_lidar_full`	`identity_li`
`capacity`	2.5	1.6	TBC	4.0
`learning_rate`	0.025	0.025	TBC	0.05
`max_iter` (derived)	100	64	TBC	80
`max_leaf_nodes`	16	31	TBC	8
`min_samples_leaf`	5	5	TBC	5
`max_bins`	255	255	TBC	255

Parameter	Selected	Best PR-AUC (climate)	Best PR-AUC (full)	Best QWK
`l2_regularization`	0.0	0.0	TBC	1.0
PR-AUC(4)	0.1966	0.1977	0.1960	0.1954
QWK	0.2796	0.2802	0.2845	0.2883

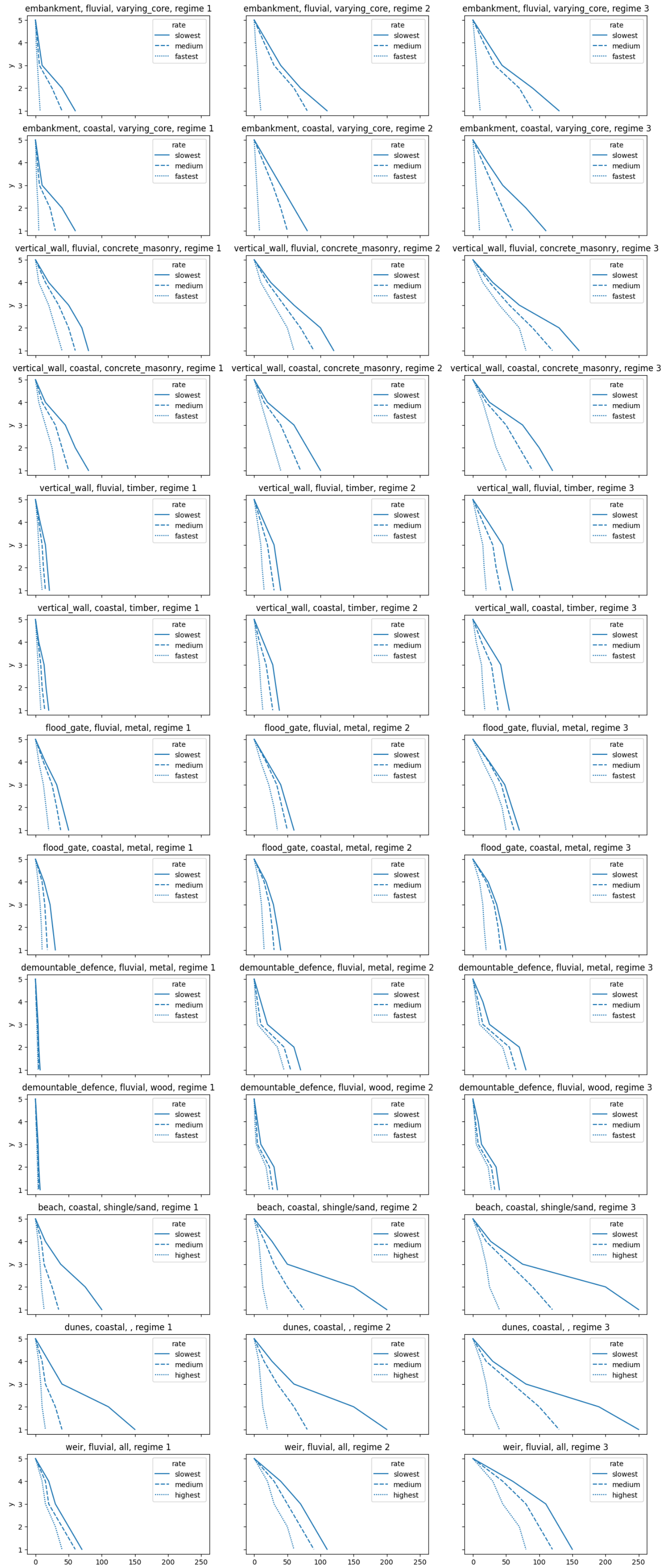

Figure C.2: Full Heuristic Curves

Appendix D

Ethics Approval Form

Table D.1: Ethical Approval Form

Section1: Students Details	...2
Student Record ID	14153
Faculty	College of Arts, Technology and Environment
Department	School of Computing and Creative Technologies
Supervisor First Name	Kat
Supervisor Last Name	Fair
I confirm that I have completed the Research Data Management Training	Yes
Project Details Proposed Project Title	“Does a machine learning approach to predicting flood defence condition grade improve on established heuristic baselines?”
Level of Study	PGT
Faculty	College of Arts, Technology and Environment
Department	School of Computing and Creative Technologies
Module Code	UFCF9Y-60-M
Module Name	CSCT Masters Project
Module Leader	Rachel Long
Student Name	Daniel Hill (Student)
Student Number	24057087

Table D.1: Ethical Approval Form *(continued)*

Section1: Students Details	...2
Project Details - Record Questions A.	NA
Are human participants involved?	No
Have you seen and approved the UWE participant information sheet?	No
Have you seen and approved the UWE consent form?	Please select one
Have you seen and approved the research instrument (e.g.questionnaire/interview schedule)?	Please select one
B. Does the proposed research fall into any of the following categories?	NA
Research involving potentially vulnerable groups:(*except for educational research in classroom settings where safeguarding arrangements are already in place, and where appropriate parent and pupil consent for the proposed UWE research activity is also in place), (*N.B. Applicable to UWE Education Department students only), people with a learning disability or cognitive impairment, those who lack decision making capacity or people in a dependent or unequal relationship?	Select one
Research involving NHS or independent hospital patients, social care (including care/nursing homes) service users or prisoners?	Select one
Research involving deception or which is conducted without participants' full and informed consent at the time the study is carried out, for example studies using data from social media	Select one
Research involving the collection of data, or access to data/ records,involving personal or sensitive confidential information, including genetic or other biological information concerning identifiable individual or Special Category Data under the Data Protection Act, or linkage of datasets with the result that individuals can be identified	Select one
Research which would or might induce psychological stress, anxiety or humiliation, or cause more than minimal pain or distress to either participants or researchers	Select one
Research involving intrusive interventions or data collection methods,e.g. the administration of substances, taking of biological samples, vigorous physical exercise or techniques such as hypnotism. This includes where participants are persuaded to reveal information they would not otherwise disclose in the course of their everyday lives or within public forums	Select one
Research involving visual/vocal methods, where participants or other individuals may be identifiable in the visual images used or generated	Select one
Research which may involve data sharing of confidential information beyond the initial consent given, e.g. where the research topic or data gathering involve a risk of information being disclosed that would require researchers to breach confidentiality conditions agreed with participants	Select one

Table D.1: Ethical Approval Form *(continued)*

Section1: Students Details	...2
Research on, or which may elicit information about, sensitive topics,including but not limited to sex and sexuality, security sensitive information including extreme beliefs or views, experiences of abuse or harm, religious beliefs, drug use, criminal activity, ethnicity/racism	Select one
Special Categories	NA
Research involving human tissue, including body parts blood, and cells	No
Research using administrative data not in the public domain, secure data or security-sensitive data	No
NA	NA
Research where data collection is undertaken or shared outside the UK,or where personal data will cross National boundaries	No
NA	NA
Research involving animals (including invertebrates) or animal by-products	No
NA	NA
Research which might have a negative environmental impact	No
NA	NA
Research involving politically and/or culturally sensitive funding sources or partners	No
Research involving financial inducements (other than reasonable expenses and compensation for time)	No